

PRAKTIKUM SISTEM OPERASI

**MODUL IX
(SYSCALL)**



**Universitas
Telkom**

Oleh:

(Naura Aisha Zahira)

(2311104078)

PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

UNIVERSITAS TELKOM

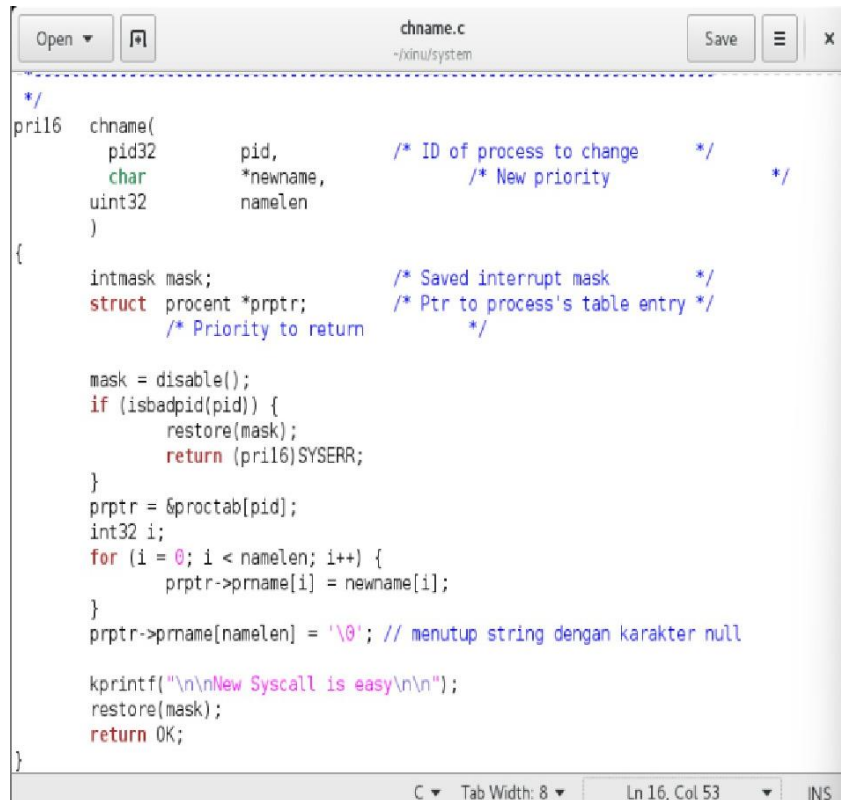
2026

Jurnal

1. Buat syscall baru seperti yang ditunjukkan pada modul syscall poin 9.5!

Jawab:

- Kode pada xinu/system/chname.c

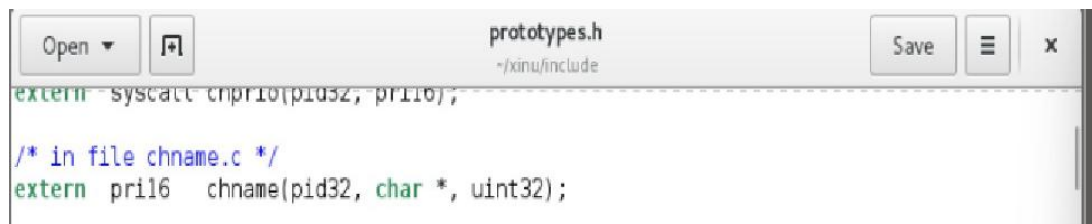


```
*/
pri16 chname(
    pid32    pid,          /* ID of process to change */
    char     *newname,     /* New priority */
    uint32    namelen
)
{
    intmask mask;          /* Saved interrupt mask */
    struct proctab *prptr; /* Ptr to process's table entry */
    /* Priority to return */

    mask = disable();
    if (isbadpid(pid)) {
        restore(mask);
        return (pri16)SYSERR;
    }
    prptr = &proctab[pid];
    int32 i;
    for (i = 0; i < namelen; i++) {
        prptr->pname[i] = newname[i];
    }
    prptr->pname[namelen] = '\0'; // menutup string dengan karakter null

    kprintf("\n\nNew Syscall is easy\n\n");
    restore(mask);
    return OK;
}
```

- Kode pada xinu/include/prototypes.h



```
extern syscall chprio(pid32, pri16);

/* in file chname.c */
extern pri16 chname(pid32, char *, uint32);
```

- hasil



```
Welcome to Xinu!

xsh $ uptime
Xinu has been up 13 second(s)
xsh $
```

2. Perbaiki syscall chprio (xinu/system/chprio.c) dengan memperhatikan validasi input -

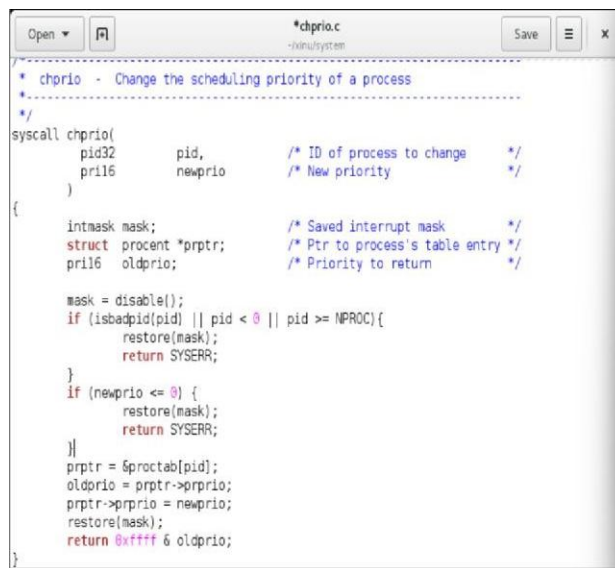
- Pastikan id adalah angka dari 0 – NPROC (ukuran maks banyaknya proses)
- Pastikan prioritas adalah bilangan yang positif

Compile dan jalankan Xinu dengan syscall yang telah diperbaiki

- make clean
- make

Jawab:

- kode pada xinu/system/chprio.c



```
* chprio - Change the scheduling priority of a process
*.....
*/
syscall chprio(
    pid32    pid,      /* ID of process to change */
    pri16    newprio   /* New priority */
)
{
    intmask mask;      /* Saved interrupt mask */
    struct procent *prptr; /* Ptr to process's table entry */
    pri16 oldprio;      /* Priority to return */

    mask = disable();
    if (isbadpid(pid) || pid < 0 || pid >= NPROC){
        restore(mask);
        return SYSERR;
    }
    if (newprio <= 0){
        restore(mask);
        return SYSERR;
    }
    prptr = &proctab[pid];
    oldprio = prptr->pprio;
    prptr->pprio = newprio;
    restore(mask);
    return 0xffff & oldprio;
}
```

- make clean

```
xinu@xinu-develop-end:~/xinu/compile$ make clean
removing .o files
removing configuration files ...
make[1]: Entering directory '/home/xinu/xinu/config'
rm -f config lex.yy.c y.tab.c
make[1]: Leaving directory '/home/xinu/xinu/config'
removing xinu ...
xinu@xinu-develop-end:~/xinu/compile$
```

- make

```
xinu@xinu-develop-end: ~/xinu/compile
File Edit View Search Terminal Help
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\"cat version\" -I../i
nclude -o binaries/xsh_udpecho.o ../shell/xsh_udpecho.c
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\"cat version\" -I../i
nclude -o binaries/xsh_udpsvr.o ../shell/xsh_udpsvr.c
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\"cat version\" -I../i
nclude -o binaries/xsh_uptime.o ../shell/xsh_uptime.c
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\"cat version\" -I../i
nclude -o binaries/clkdsp.o ../system/clkdsp.S
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\"cat version\" -I../i
nclude -o binaries/ctxsw.o ../system/ctxsw.S
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\"cat version\" -I../i
nclude -o binaries/intr.o ../system/intr.S
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\"cat version\" -I../i
nclude -o binaries/ttydispatch.o ../device/tty/ttydispatch.S
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\"cat version\" -I../i
nclude -o binaries/ethdispatch.o ../device/eth/ethdispatch.S

Loading object files to produce GRUB bootable xinu
Copying xinu.elf to xinu.boot in the TFTP directory.

xinu@xinu-develop-end:~/xinu/compile$
```

3. Lakukan hal-hal berikut ini

- Edit xsh_uptime.c Tambahkan kode berikut

```
/* Check for valid number of arguments */
if (nargs > 1) {
    fprintf(stderr, "%s: too many arguments\n", args[0]);
    fprintf(stderr, "Try '%s --help' for more information\n",
            args[0]);
    return 1;
}

// tambahan code
chprio(5,33) // memanggil syscall chprio; mengubah prioritas menjadi 33 pada id=5
```

- Compile source code tersebut dengan perintah

- make clean

```
xinu@xinu-develop-end:~/xinu/compile$ make clean
removing .o files
removing configuration files ...
make[1]: Entering directory '/home/xinu/xinu/config'
rm -f config lex.yy.c y.tab.c
make[1]: Leaving directory '/home/xinu/xinu/config'
removing xinu ...
xinu@xinu-develop-end:~/xinu/compile$
```

- make

```
xinu@xinu-develop-end: ~/xinu/compile
File Edit View Search Terminal Tabs Help

xinu@xinu-develop-end: ~/xinu/shell  x  xinu@xinu-develop-end: ~/xinu/compile  x

-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=""`cat version`"\ -I../i
nclude -o binaries/xsh_udpsvr.o ../shell/xsh_udpsvr.c
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=""`cat version`"\ -I../i
nclude -o binaries/xsh_uptime.o ../shell/xsh_uptime.c
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=""`cat version`"\ -I../i
nclude -o binaries/clkdisp.o ../system/clkdisp.S
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=""`cat version`"\ -I../i
nclude -o binaries/ctxsw.o ../system/ctxsw.S
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=""`cat version`"\ -I../i
nclude -o binaries/intr.o ../system/intr.S
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=""`cat version`"\ -I../i
nclude -o binaries/ttydispatch.o ../device/tty/ttydispatch.S
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=""`cat version`"\ -I../i
nclude -o binaries/ethdispatch.o ../device/eth/ethdispatch.S

Loading object files to produce GRUB bootable xinu
Copying xinu.elf to xinu.boot in the TFTP directory.

xinu@xinu-develop-end:~/xinu/compile$ sudo min
```

- Jalankan perintah ps

- xsh \$ ps

- perhatikan prioritas proses dengan id = 5

```
xsh $ ps
```

Pid	Name	State	Prio	Ppid	Stack	Base	Stack	Ptr	Stack	Size	Msg	Content
0	prnull	ready	0	0	0x005FDFFC	0x00FFFED8	8192	1	3			
1	ipout	wait	500	0	0x005FBFFC	0x005FBF30	8192	0	0			
2	netin	wait	500	0	0x005F9FFC	0x005F9ED0	8192	0	0			
4	Main process	recv	20	3	0x005E7FFC	0x005E7F34	65536	0	0			
5	shell	recv	50	4	0x005F7FFC	0x005F79AC	8192	0	0			
6	ps	curr	20	5	0x005F5FFC	0x005F5FC4	8192	0	0			

- Jalankan uptime
 - xsh \$ uptime
 - Perhatikan hasil perintah tersebut

```
xsh $ uptime
uptime: 7 seconds
```

- Jalankan ps
 - xsh \$ ps
 - perhatikan prioritas proses dengan id = 5 seharusnya sudah berubah

```
xsh $ ps
Pid Name          State Prio Ppid Stack Base Stack Ptr  Stack Size Msg Content
-----
0 prnull          ready  0    0 0x005FDFFC 0x00FFFECC      8192  1      3
1 ipout           wait   500  0 0x005FBFFC 0x005FBF30      8192  0      0
2 netin           wait   500  0 0x005F9FFC 0x005F9ED0      8192  0      0
4 Main process    recv   20   3 0x005E7FFC 0x005E7F34     65536  0      0
5 shell           recv   33   4 0x005F7FFC 0x005F79AC      8192  0      7
8 ps              curr   20   5 0x005F5FFC 0x005F5FC4      8192  0      0
xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:0 | ttyS0
```

Testing chprio syscall yang telah diubah

- Testing prioritas tidak boleh < 0: Ubah “chprio(5,33)” menjadi “chprio(5,-3)” pada xsh_uptime.c

```

        args[0]);
    return 1;
}
// tambahan kode
chprio(5,-3);
secs = clktime; /* Total seconds since boot */

printf("uptime: %u seconds\n", secs);
return 0;
}

```

```
xsh $ ps
Pid Name          State Prio Ppid Stack Base Stack Ptr  Stack Size Msg Content
-----
0 pmull           ready  0    0 0x005FDFFC 0x00FFFED8 8192 1 3
1 ipout           wait  500  0 0x005FBFFC 0x005FBF30 8192 0 0
2 netin           wait  500  0 0x005F9FFC 0x005F9ED0 8192 0 0
4 Main process    recv  20   3 0x005E7FFC 0x005E7F34 65536 0 0
5 shell           recv  50   4 0x005F7FFC 0x005F79AC 8192 0 0
6 ps              curr  20   5 0x005F5FFC 0x005F5FC4 8192 0 0

xsh $ uptime
uptime: 33 seconds

xsh $ ps
Pid Name          State Prio Ppid Stack Base Stack Ptr  Stack Size Msg Content
-----
0 pmull           ready  0    0 0x005FDFFC 0x00FFFED8 8192 1 3
1 ipout           wait  500  0 0x005FBFFC 0x005FBF30 8192 0 0
2 netin           wait  500  0 0x005F9FFC 0x005F9ED0 8192 0 0
4 Main process    recv  20   3 0x005E7FFC 0x005E7F34 65536 0 0
5 shell           recv  33   4 0x005F7FFC 0x005F79AC 8192 0 7
8 ps              curr  20   5 0x005F5FFC 0x005F5FC4 8192 0 0

xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:2 | ttyS0
```

- Testing id adalah valid: Ubah “chprio(5,33)” menjadi “chprio(3000,3)”

```
        return 1;
    }
    // tambahan kode
    chprio(3000,3);
    secs = clktme; /* Total seconds since boot */

    printf("uptime: %u seconds\n", secs);
    return 0;
}
```



```

xsh $ ps
Pid Name                State Prio Ppid Stack Base Stack Ptr  Stack Size Msg Content
-----
0 pnull                 ready  0    0 0x005FDFFC 0x00FFFED8 8192 1 3
1 ipout                 wait  500  0 0x005FBFFC 0x005FBF30 8192 0 0
2 netin                wait  500  0 0x005F9FFC 0x005F9ED0 8192 0 0
4 Main process         recv  20   3 0x005E7FFC 0x005E7F34 65536 0 0
5 shell                recv  50   4 0x005F7FFC 0x005F79AC 8192 0 0
6 ps                   curr  20   5 0x005F5FFC 0x005F5FC4 8192 0 0

xsh $ uptime
uptime: 5 seconds

xsh $ ps
Pid Name                State Prio Ppid Stack Base Stack Ptr  Stack Size Msg Content
-----
0 pnull                 ready  0    0 0x005FDFFC 0x00FFFED8 8192 1 3
1 ipout                 wait  500  0 0x005FBFFC 0x005FBF30 8192 0 0
2 netin                wait  500  0 0x005F9FFC 0x005F9ED0 8192 0 0
4 Main process         recv  20   3 0x005E7FFC 0x005E7F34 65536 0 0
5 shell                recv  50   4 0x005F7FFC 0x005F79AC 8192 0 7
8 ps                   curr  20   5 0x005F5FFC 0x005F5FC4 8192 0 0

xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:0 | ttyS0

```

Hasil dua testing di atas adalah prioritas tidak berubah karena salah argument
(dibuktikan dengan menggunakan perintah ps)